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Vision-Based Intelligent Gait Environment Detection Algorithm for Assisting Stroke Patients with a Soft Ankle Exosuit

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Intelligent Robotics
Laboratory

I. Introduction

II. Soft Exosuit

1. Soft Exosuit Prototype
2. Control Strategy

III. Environment Recognition System

IV. Experiments

V. Results

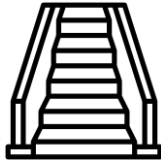
1. Real-Time Environment Classification Performance
2. Joint Kinematics
3. Muscle Reduction

VI. Conclusion

I. Introduction

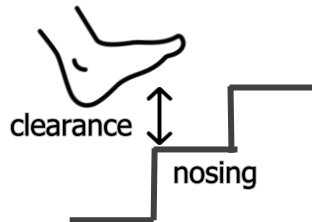
Stroke Patients

- functional impairments & gait disturbances
- **foot drop** : reduction in foot clearance (swing phase of gait cycle)



stair ambulation

- common activity
- challenging task : joint range, muscular moment $\uparrow$
- ascent: insufficient clearance between foot & stair nosing



Lower Limb Exoskeleton

- assist gait for stair environment

I. Introduction

Exoskeleton

inertial measurement unit (IMU) & insole sensor

- provide current state of gait
- Limitation: Lack of prior environmental information

vision – based perception



human
peripheral vision

- ✓ visual recognition of step location
- ✓ continuous monitoring stage



vision system
for exoskeleton

- ✓ awareness of surrounding environment
- ✓ locomotion guidance

I. Introduction

environment classification + exoskeleton

[challenge]

- transition phase accuracy
- stroke patient consideration

Deep learning-based methodologies

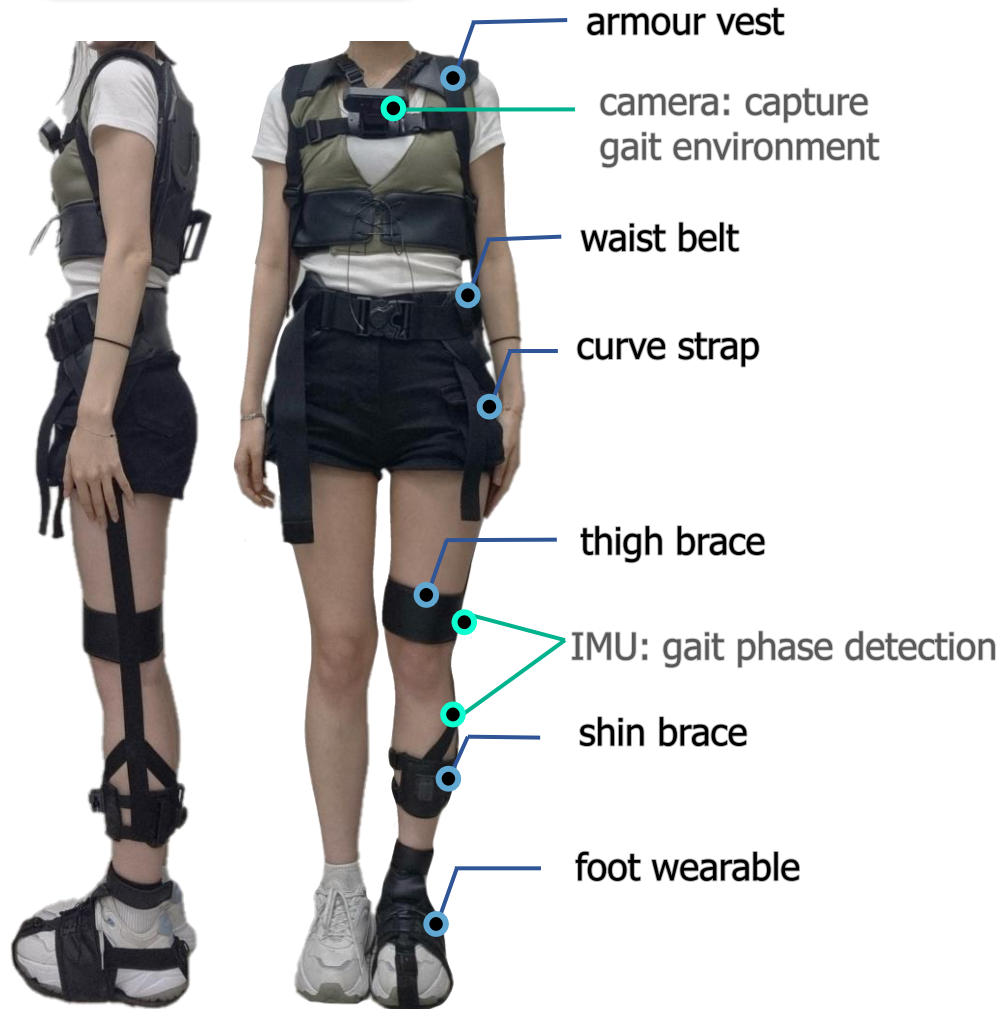
influence robotics field

Vision-based Intelligent gait environment detection algorithm for ankle exosuit

- Integration of deep learning-based environment classification into exoskeleton application
- detailed gait environment classification (YOLOv8) + spatio-temporal feature extraction (LSTM)
- soft ankle exosuit aimed at stroke patients

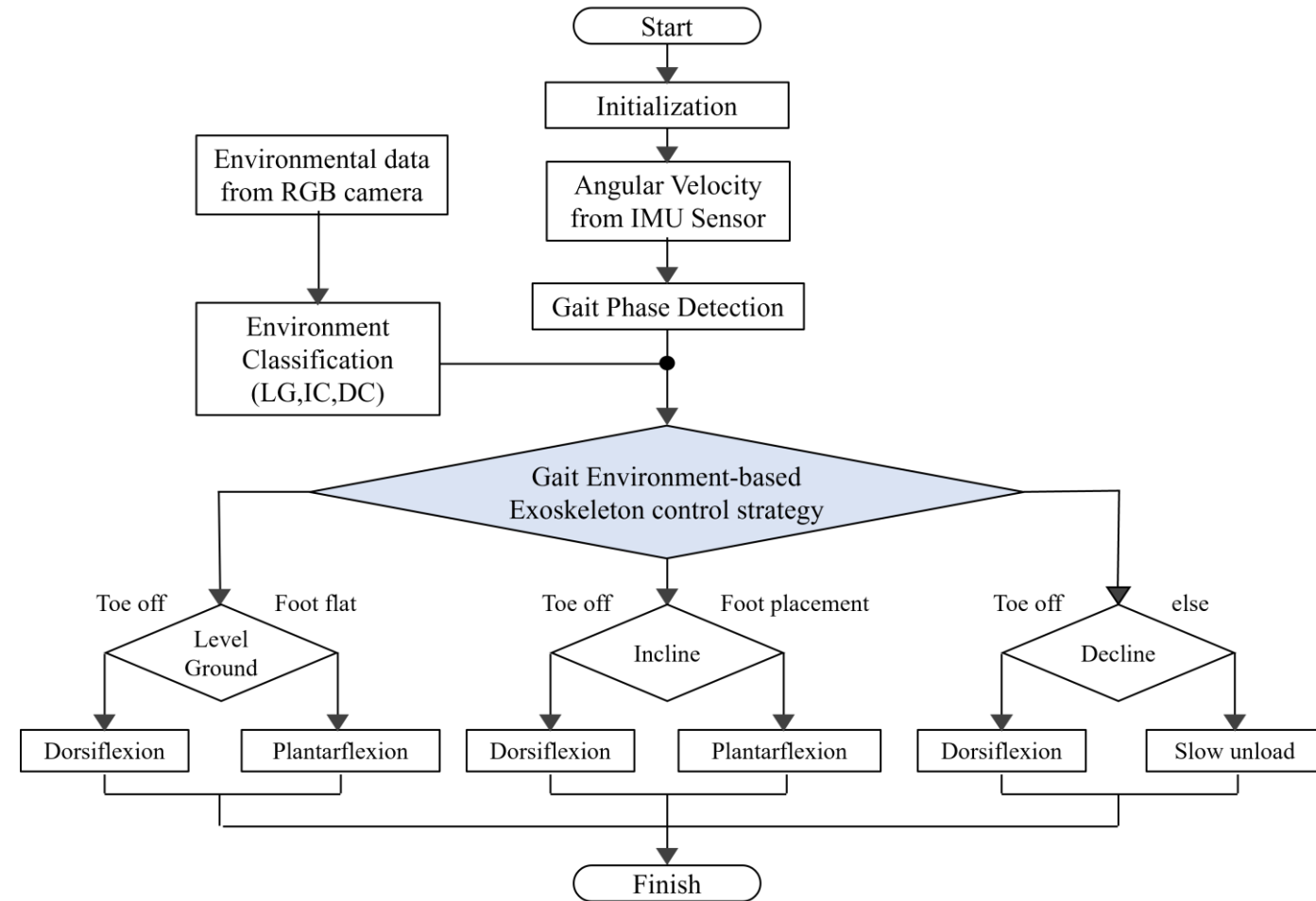
II . Soft Exosuit

Exosuit Design



- flexible and lightweight materials
- Bowden cable-driven method

Control Strategy



flowchart of gait-environment-based exoskeleton control strategy

III . Environment Recognition System

Data Acquisition · Class Definition

[customized dataset]

- level ground and stair environment
- 960 x 540 pixel
- training: 16,000 | valid: 2,000 | test: 2000

-> 8:1:1

train

valid

test



IC prev

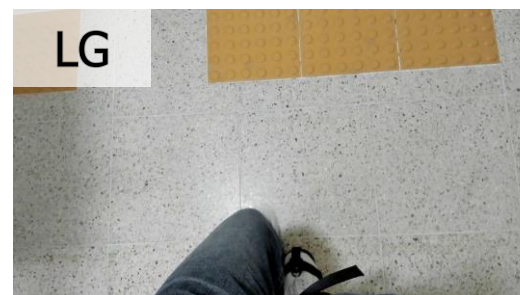
initial incline

IC

mid-incline

IC next

final incline



LG

level ground



DC prev

initial decline

DC

mid-decline

DC next

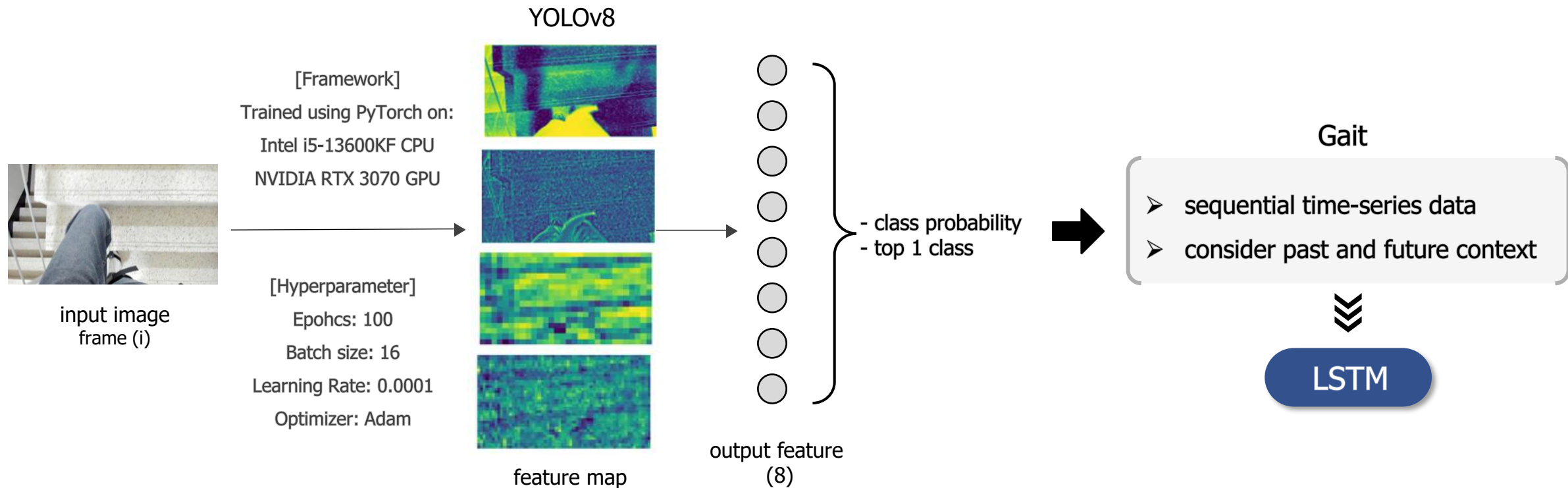
final decline

III . Environment Recognition System

YOLOv8 Environment Classification

- object detection framework: one-stage detection approach
- provide real-time contextual insight for frame

{ Rapid Detection: critical for detecting gait environment transitions
Spatial analysis: Effective in analyzing global spatial information



III . Environment Recognition System

LSTM-Based Spatio-Temporal Feature Extraction

- extract temporal features: adapting to environment transitions



Data Preparation

- Frame Sampling: **20 FPS**
- 1 class for 10 frames,
with a stride of 5 (**4 labels/sec**)
- Sequence shape: **(1714, 10,8)**



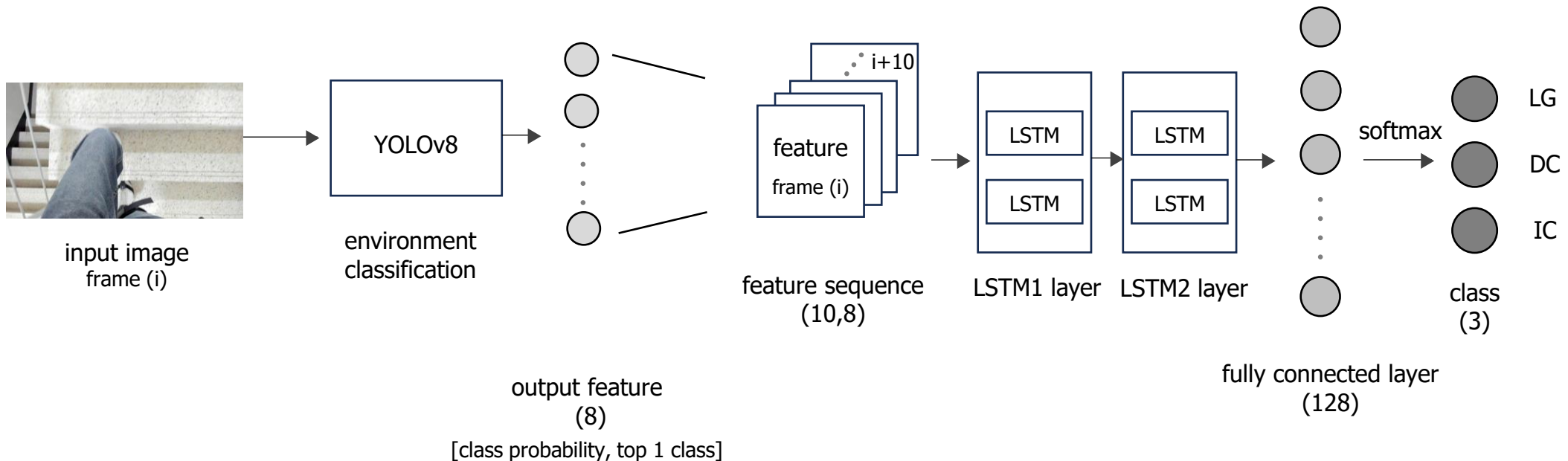
Class Labeling

- **Most probable Class:**
Chosen from YOLO frame-wise probabilities
- **Maximum Voting & Average Voting**



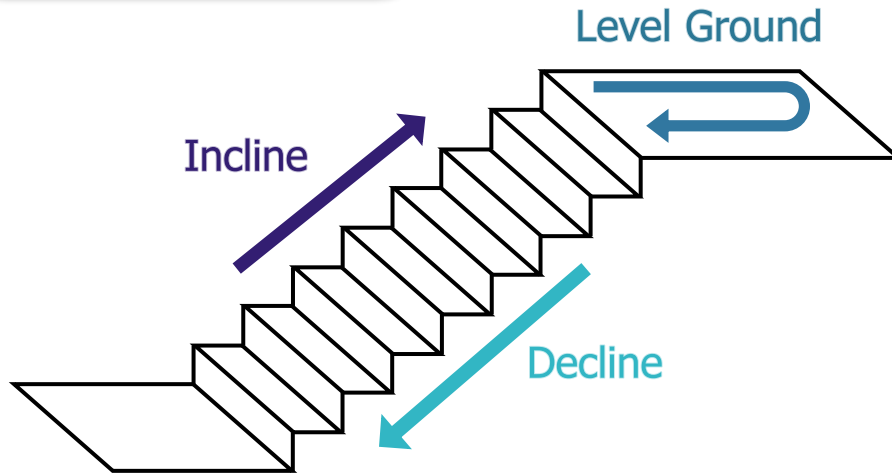
Model Training

- 1000 epochs
- batch sizes : 32
- learning rate: 0.001
- Adam optimizer



IV . Experiments

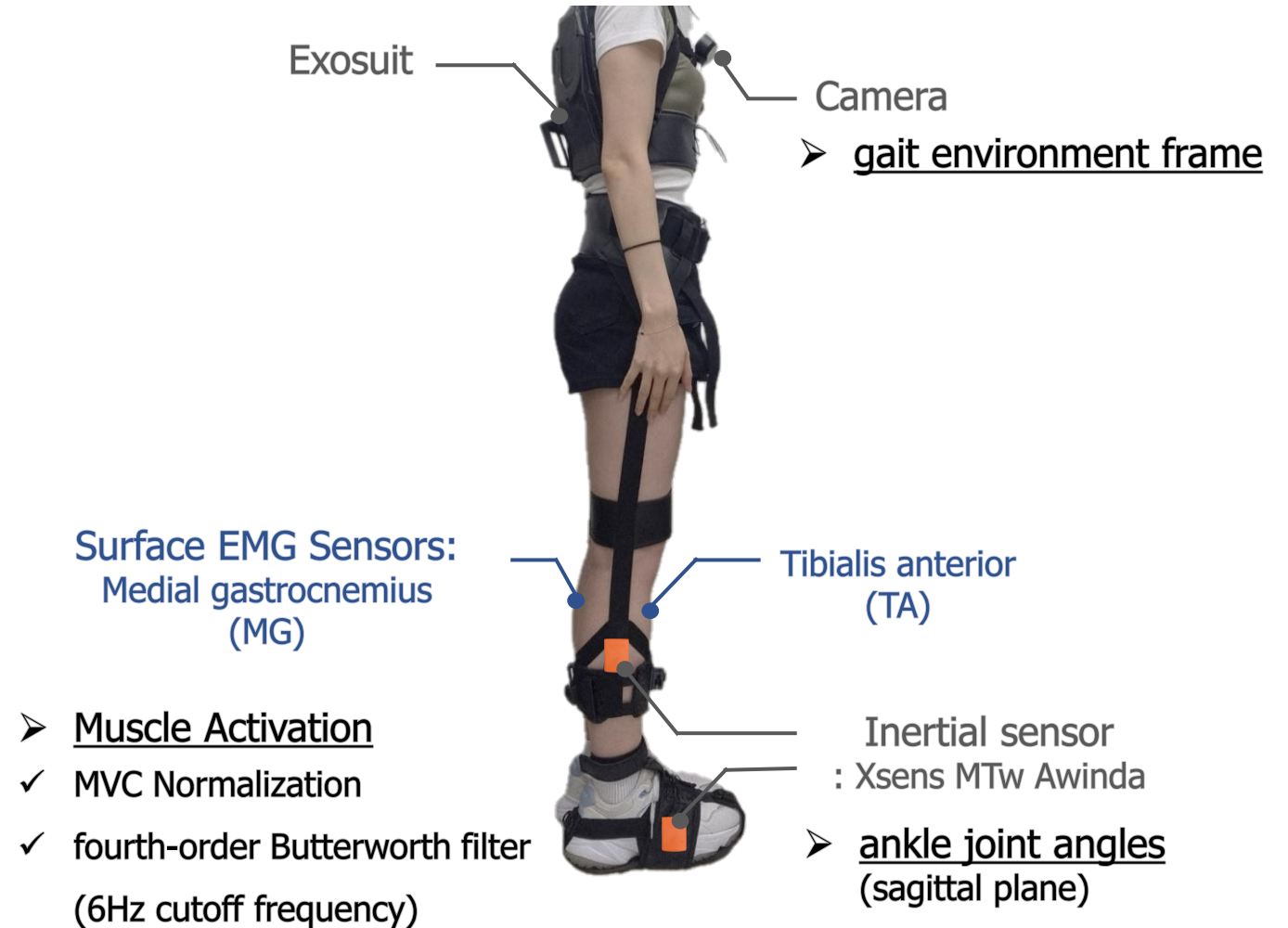
Experimental Setup



Condition

- 1) without exosuit (*No Exo*)
- 2) Exosuit worn but unpowered (*Exo Off*)
- 3) Exosuit worn and powered (*Exo On*)

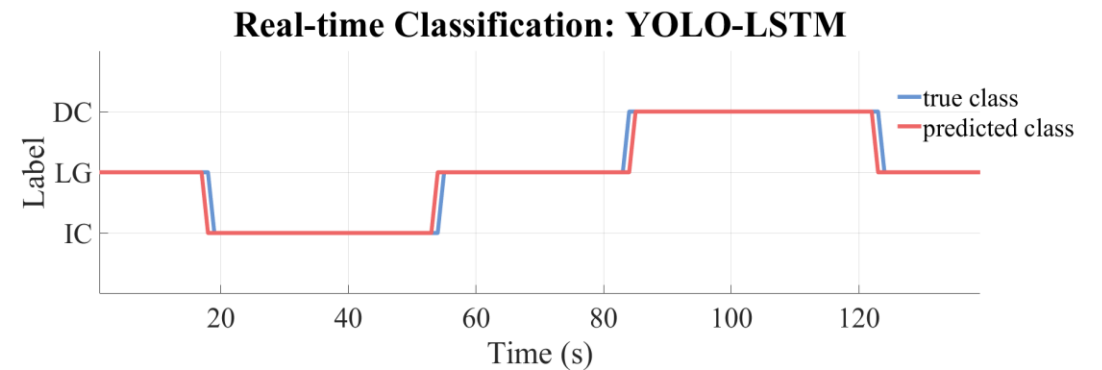
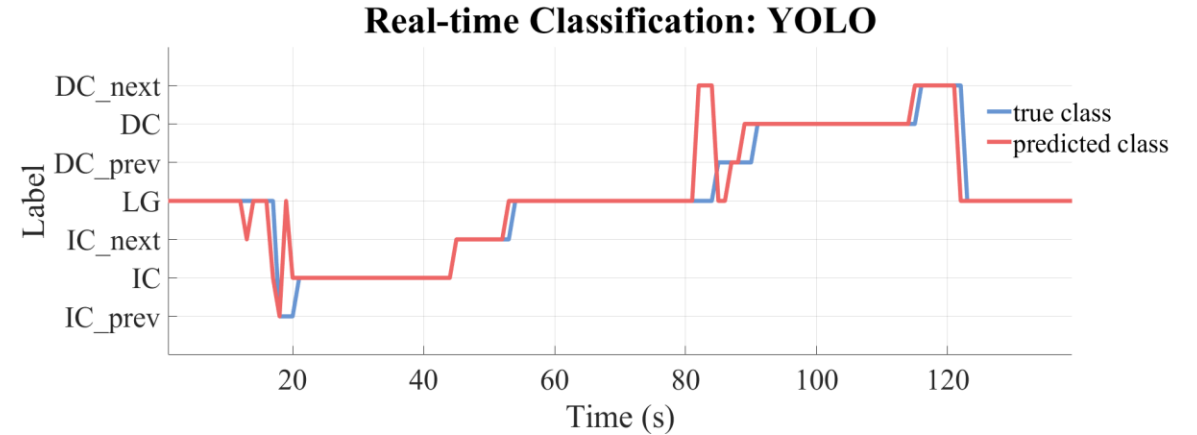
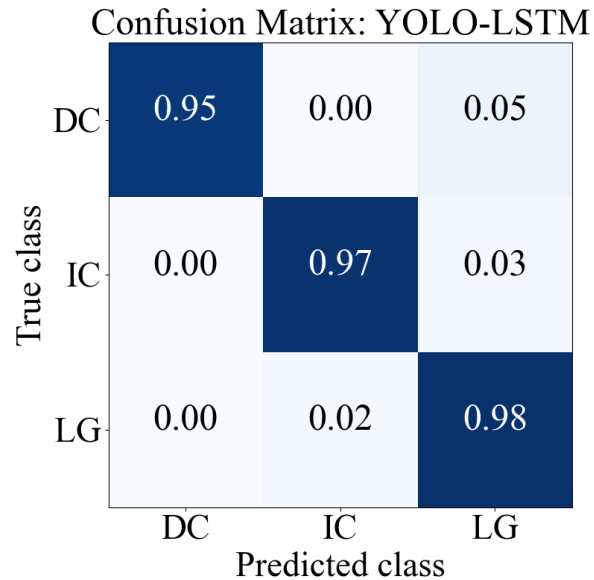
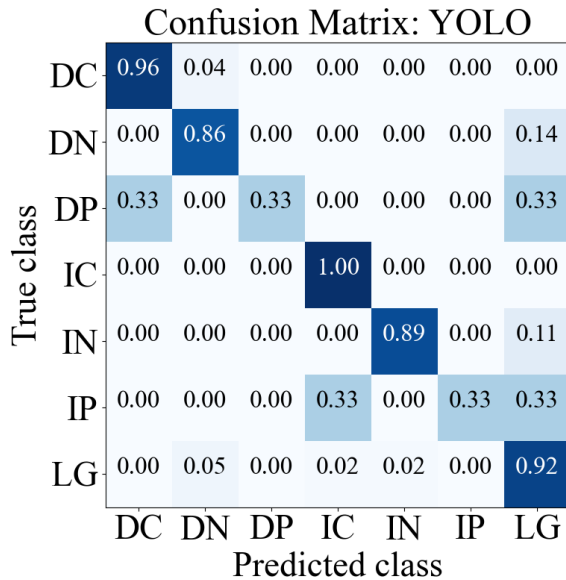
10-minute familiarization with exosuit
20-minute rest between conditions



- Muscle Activation
- ✓ MVC Normalization
- ✓ fourth-order Butterworth filter
(6Hz cutoff frequency)

V . Results

Real-Time Environment Classification Performance



YOLO

Low Recall in Transition Classes (IC_{prev} , DC_{prev})

Overall F1 score : **0.774** - transition misclassification

YOLO - LSTM

Overall F1 score : **0.971** - balanced classification performance

YOLO

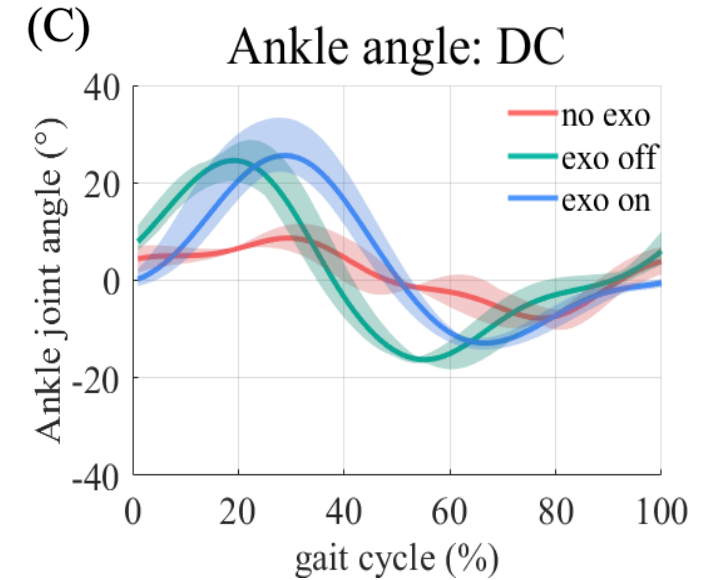
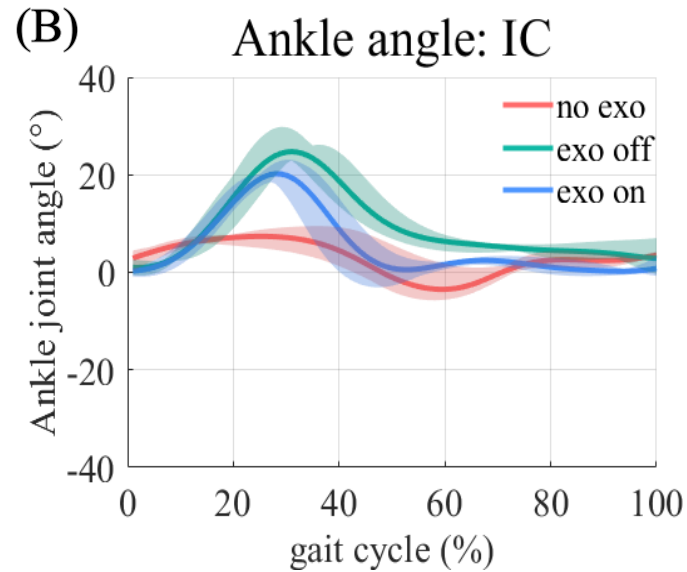
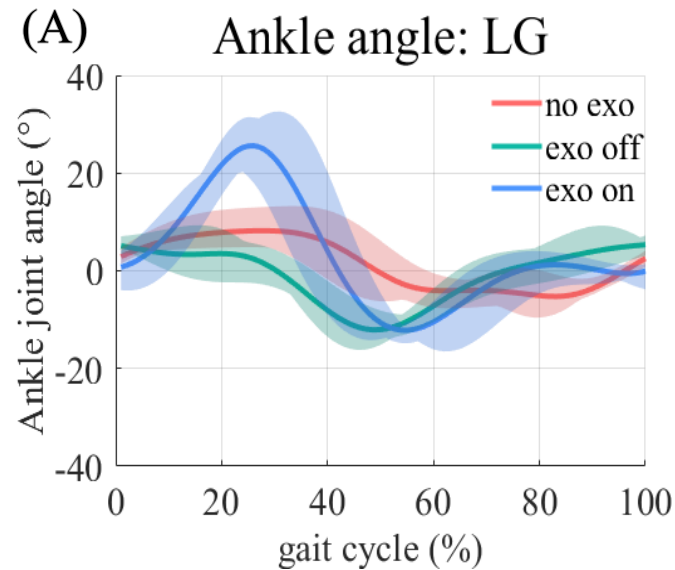
Misclassified LG as stairs 1.5s (6 frames) before stair transition

YOLO - LSTM

rapid and accurate classification during the transition phase

V . Results

Joint Kinematics



peak ankle joint dorsiflexion angle

	LG	IC	DC
<u>no exo</u>	8.88° ± 3.95°	7.92° ± 1.48°	9.14° ± 2.27°
<u>exo off</u>	6.74° ± 2.08°	25.99° ± 2.78°	25.08° ± 3.63°
<u>exo on</u>	27.13° ± 5.71°	21.06° ± 2.08°	25.65° ± 5.16°

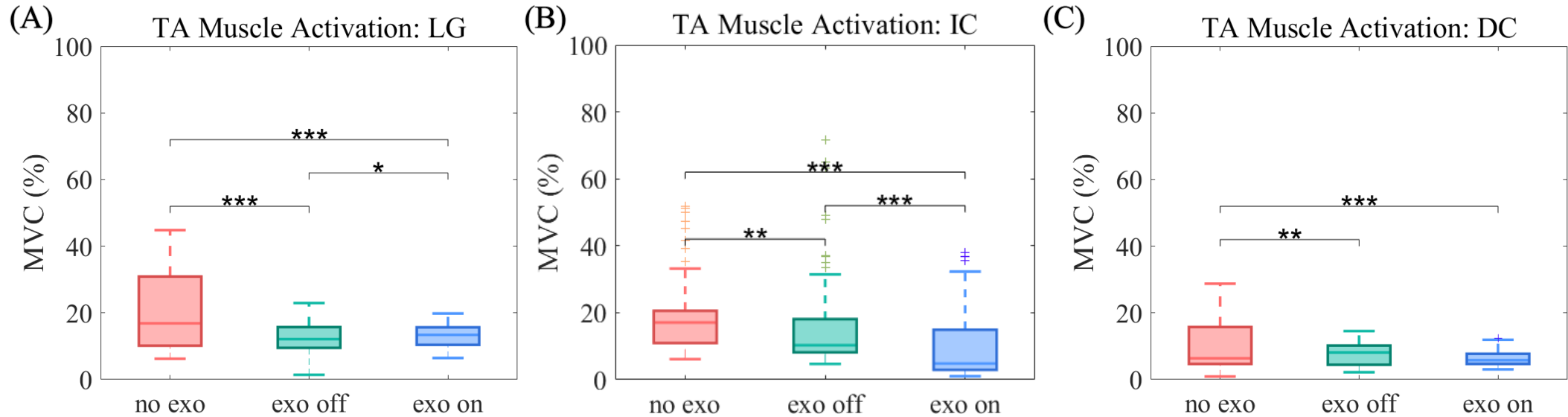


LG: **significant increase** with the *exo on*

IC: **increase** with the *exo on*, *exo off*

V . Results

Muscle Reduction



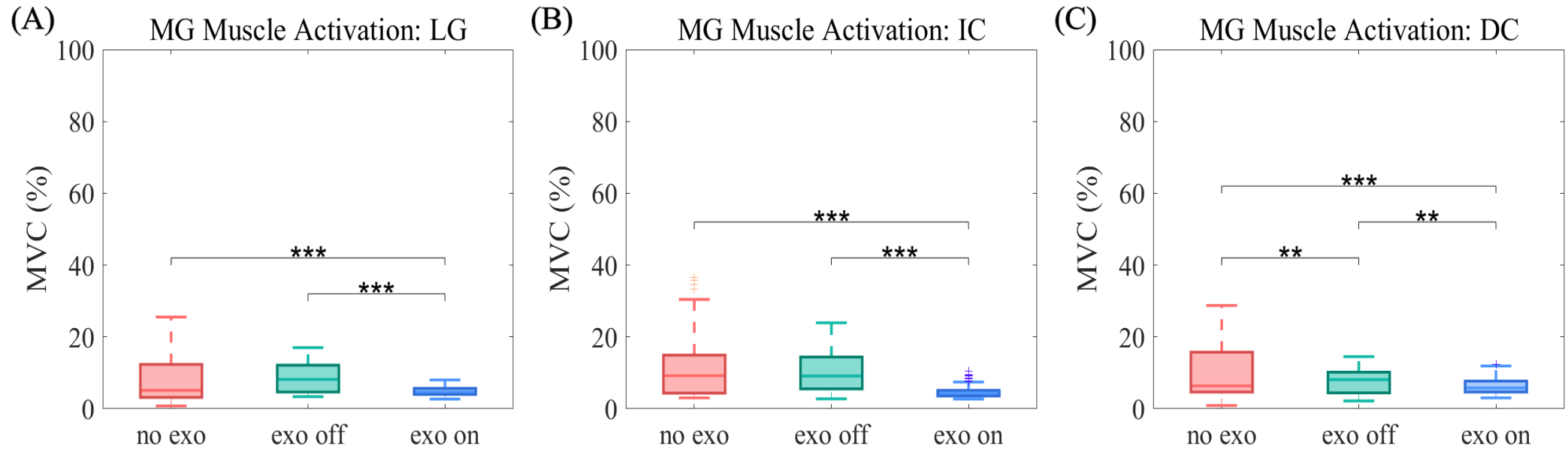
average TA muscle activation

	LG	IC	DC
<u>no exo</u>	20.63 ± 11.56%	18.44% ± 9.92%	10.75% ± 4.25%
<u>exo off</u>	12.77% ± 5.02%	15.24% ± 12.77%	12.34% ± 10.93%
<u>exo on</u>	13.22% ± 3.43%	9.79% ± 10.12%	12.09% ± 4.81%

LG, IC: **reduction** in *exo off*, *exo on* condition
DC: slight increases with exosuit

V . Results

Muscle Reduction



average MG muscle activation

	LG	IC	DC
<u>no exo</u>	7.88% ± 6.61%	11.71% ± 8.83%	9.97% ± 7.35%
<u>exo off</u>	8.51% ± 4.04%	10.66% ± 5.99%	7.62% ± 3.41%
<u>exo on</u>	4.87% ± 1.21%	4.70% ± 1.69%	6.41% ± 2.19%

LG, IC, DC: **reduction** in *exo on* condition

Conclusion

- **vision-based detection algorithm** for an ankle exosuit, focusing on improving gait support for stroke patients
- integration of **YOLOv8** for environment classification and **LSTM** for spatio-temporal recognition
- **rapid and accurate classification** during gait environment transition
- **increased ankle dorsiflexion / Reduced TA and MG muscle activation**

Future work

- Validate across diverse real-world scenarios and multiple stroke patients
- Enhance the effectiveness and generalization ability of the exosuit by improving environment classification algorithm



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- ...



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Thank You

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